

Steps for Installation:

- **install kubectl search in browser**

- first tab install tool

click install on windows

The screenshot shows the Kubernetes Documentation website. The top navigation bar includes links for Kubernetes, Documentation, Kubernetes Blog, Training, Partners, Community, Case Studies, and Versions. The left sidebar contains a search bar and a list of navigation items: Documentation, Getting started, Concepts, Tasks (expanded), Install Tools (expanded), Administer a Cluster, Configure Pods and Containers, Monitoring, Logging, and Debugging, Manage Kubernetes Objects, Managing Secrets, and Inject Data Into Applications. The main content area is titled 'Install and Set Up kubectl on Windows' and includes a sub-header 'Before you begin' with a paragraph about version compatibility. Below this is the section 'Install kubectl on Windows' with a list of installation methods: 'Install kubectl binary on Windows (via direct download or curl)' and 'Install on Windows using Chocolatey, Scoop, or winget'. The page also includes a breadcrumb trail at the top: 'Kubernetes Documentation / Tasks / Install Tools / Install and Set Up kubectl on Windows'.

set path in shell:

```
C:\WINDOWS\system32> cd..
```

```
C:\WINDOWS> cd
```

```
C:\> cd kubectl
```

Paste in windows shell - :

```
C:\kubectl> curl.exe -LO
```

```
https://dl.k8s.io/release/v1.33.0/bin/windows/amd64/kubectl.exe
```

```
C:\kubectl> dir
```

Paste in shell - :

```
C:\kubectl> curl.exe -LO
```

```
https://dl.k8s.io/v1.33.0/bin/windows/amd64/kubectl.exe.sha256
```

```
C:\kubectl> CertUtil -hashfile kubectl.exe SHA256
```

```
C:\kubectl> type kubectl.exe.sha256
```

```
C:\kubectl> $(Get-FileHash -Algorithm SHA256 .\kubectl.exe).Hash -eq $(Get-Content .\kubectl.exe.sha256)
```

```
Administrator: Windows PowerShell (x86)
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> cd..
PS C:\WINDOWS> cd..
PS C:\> cd kubectl
PS C:\kubectl> curl.exe -LO "https://dl.k8s.io/release/v1.33.0/bin/windows/amd64/kubectl.exe"
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100  138  100    138    0     0   149      0  --:--:-- --:--:-- --:--:--   149
100 58.8M  100 58.8M    0     0 2596k      0  0:00:23  0:00:23 --:--:-- 3797k
PS C:\kubectl> dir

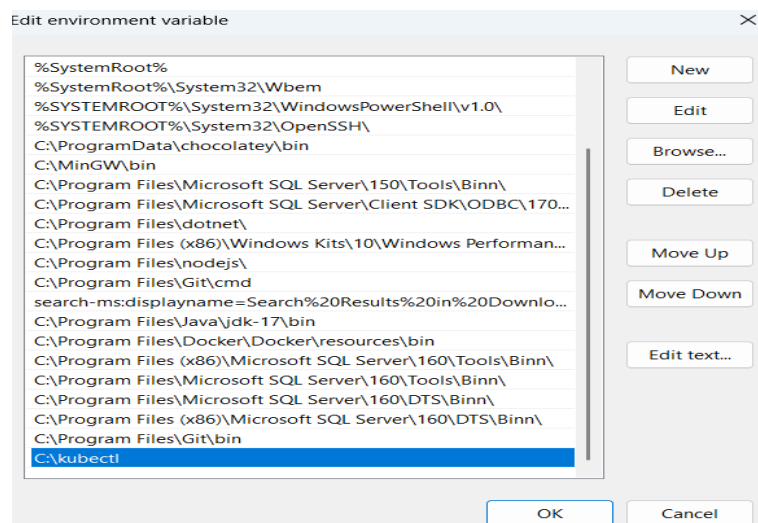
Directory: C:\kubectl

Mode                LastWriteTime         Length Name
----                -
-a----           5/15/2025   9:40 PM           61703168 kubectl.exe

PS C:\kubectl> curl.exe -LO "https://dl.k8s.io/v1.33.0/bin/windows/amd64/kubectl.exe.sha256"
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100  138  100    138    0     0   174      0  --:--:-- --:--:-- --:--:--   175
100  64  100    64    0     0    56      0  0:00:01  0:00:01 --:--:--   228
PS C:\kubectl> CertUtil -hashfile kubectl.exe SHA256
SHA256 hash of kubectl.exe:
db6d96f65a86426e6c9484ca88a233aa7f160025f40c20b153c5bf4f9746c791
CertUtil: -hashfile command completed successfully.
PS C:\kubectl> type kubectl.exe.sha256
db6d96f65a86426e6c9484ca88a233aa7f160025f40c20b153c5bf4f9746c791
PS C:\kubectl> $(Get-FileHash -Algorithm SHA256 .\kubectl.exe).Hash -eq $(Get-Content .\kubectl.exe.sha256)
True
```

Set edit environmental path

C:\kubectl

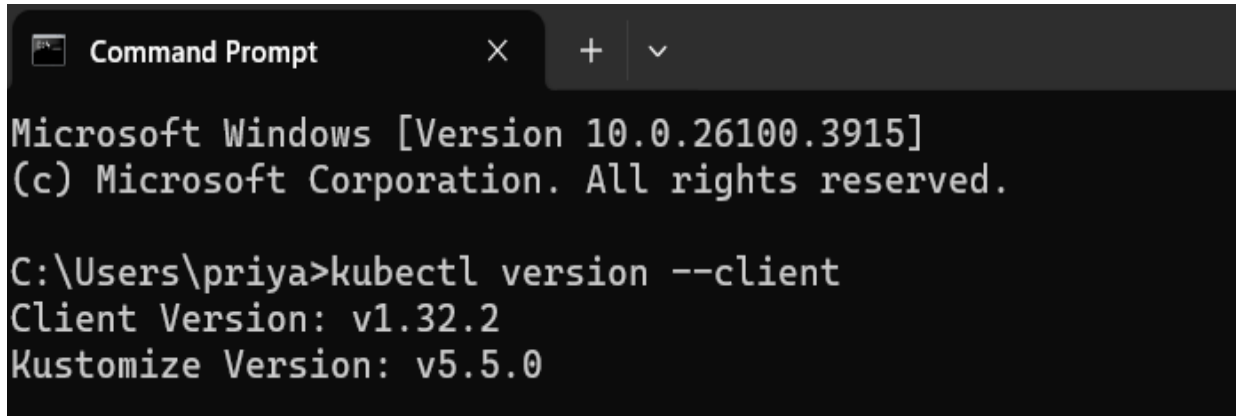


Open command prompt(cmd):-

Type command –

kubectl version --client

C:\Users\DYPATU>kubectl version --client

A screenshot of a Windows Command Prompt window. The title bar says 'Command Prompt'. The window shows the following text: 'Microsoft Windows [Version 10.0.26100.3915] (c) Microsoft Corporation. All rights reserved. C:\Users\priya>kubectl version --client Client Version: v1.32.2 Kustomize Version: v5.5.0'.

```
Microsoft Windows [Version 10.0.26100.3915]
(c) Microsoft Corporation. All rights reserved.

C:\Users\priya>kubectl version --client
Client Version: v1.32.2
Kustomize Version: v5.5.0
```

Install minikube

Search in browser

Install minikube

Create folder C:\ Minikube

Open windows shell

Set path of minikube

C:\Users\DYPATU> cd...

C:\Users> cd..

C:\> cd minikube

C:\minikube>

```
Administrator: Windows PowerShell (x86)
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> New-Item -Path 'c:\' -Name 'minikube'

-ItemType Directory -Force

Directory: C:\

Mode                LastWriteTime         Length Name
----                -
d-----          5/15/2025   9:56 PM             minikube

PS C:\WINDOWS\system32> Invoke-WebRequest -OutFile 'c:\minikube\minikube.exe' -Uri 'https://github.com/kubernetes/minikube/releases/latest/download/minikube-windows-amd64.exe' -UseBasicParsing
PS C:\WINDOWS\system32> cd..
PS C:\WINDOWS> cd..
PS C:\> cd minikube
PS C:\minikube> dir

Directory: C:\minikube

Mode                LastWriteTime         Length Name
----                -
-a----          5/15/2025   9:58 PM    125837824 minikube.exe
```

Open windows shell run as administrator:

Paste in shell

```
$oldPath = [Environment]::GetEnvironmentVariable('Path',
[EnvironmentVariableTarget]::Machine)
>> if ($oldPath.Split(';') -notcontains 'C:\minikube'){
>> [Environment]::SetEnvironmentVariable('Path', $('{0};C:\minikube' -f
$oldPath), [EnvironmentVariableTarget]::Machine)
>> }
```

```
PS C:\minikube> $oldPath = [Environment]::GetEnvironmentVariable('Path', [EnvironmentVariableTarget]::Machine)
>> if ($oldPath.Split(';') -notcontains 'C:\minikube'){
>> [Environment]::SetEnvironmentVariable('Path', $('{0};C:\minikube' -f $oldPath), [EnvironmentVariableTarget]::Machine)
>> }
```

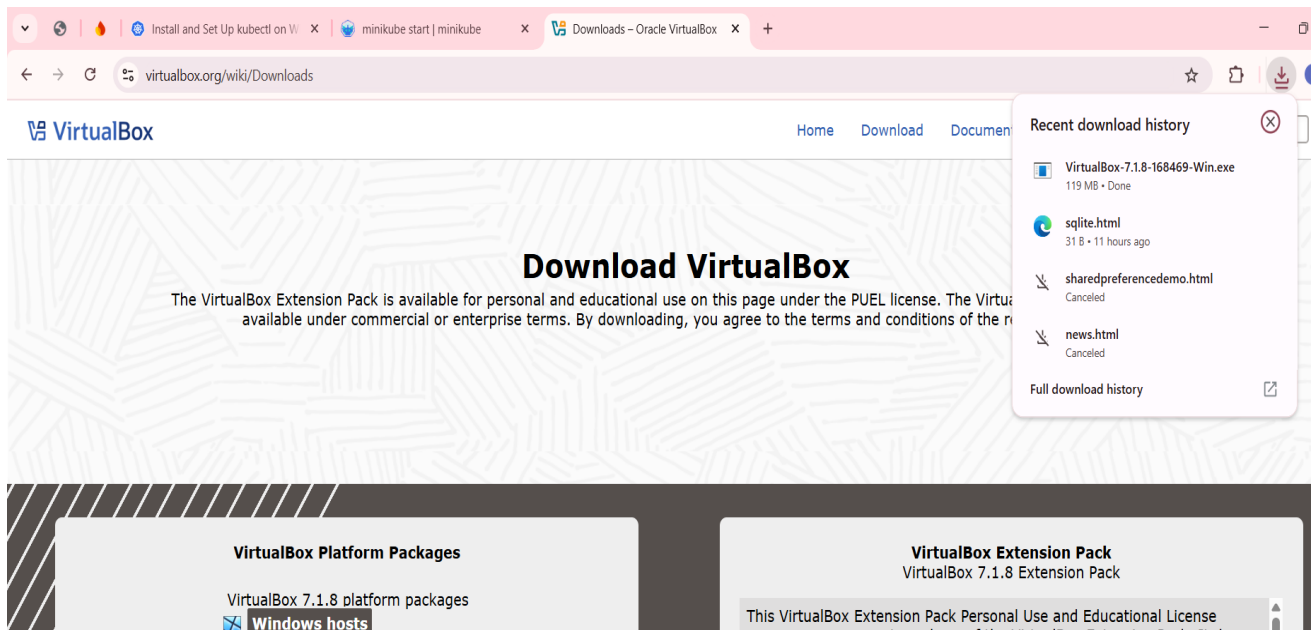
Check environmental path in minikube set

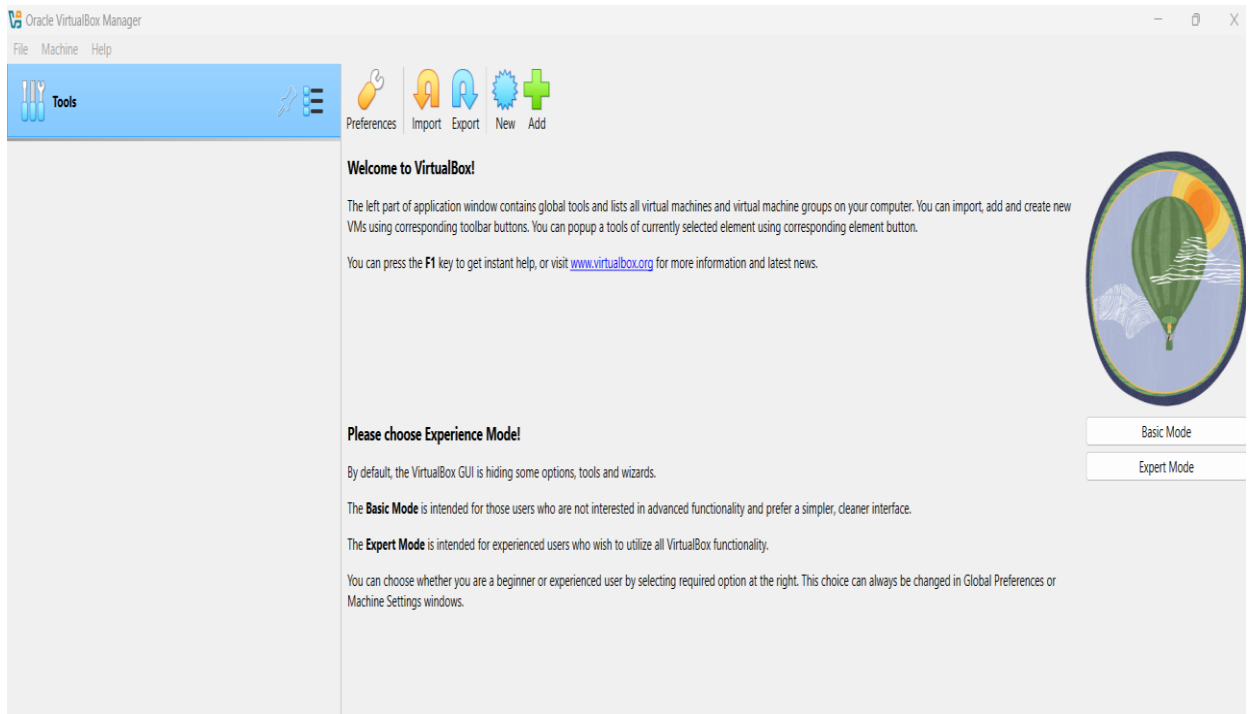
Install virtual box:

Search this link in browser- <https://www.virtualbox.org/wiki/Downloads>

Select windows hosts

Downloading virtual box is start\





Open new command prompt

```
C:\Users\DYPATU>minikube start --driver=virtualbox
```

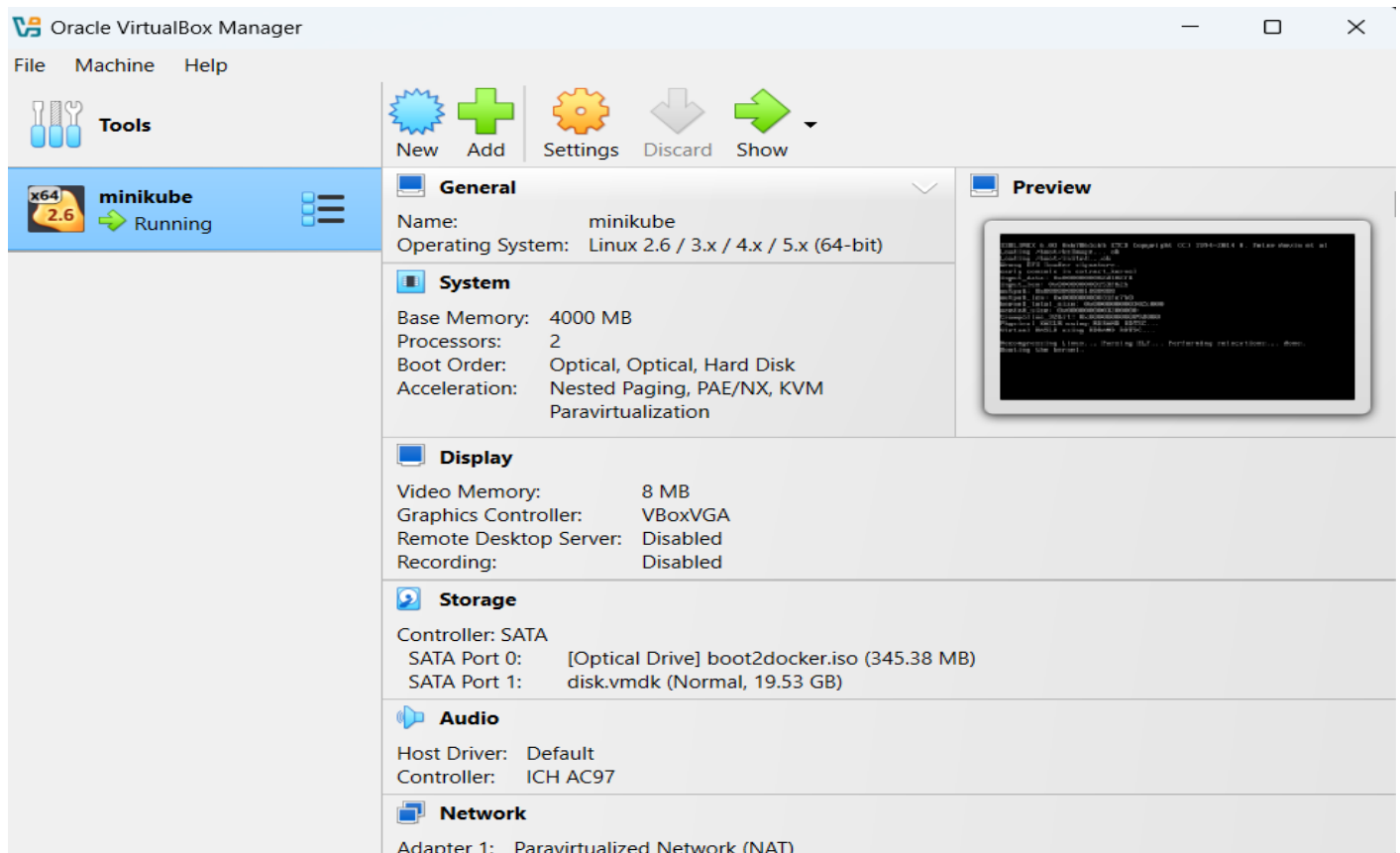
```
C:\Users\DYPATU>minikube start --no-vtx-check
```

```
Command Prompt
Microsoft Windows [Version 10.0.26100.3915]
(c) Microsoft Corporation. All rights reserved.

C:\Users\priya>minikube start --driver=virtualbox
* minikube v1.35.0 on Microsoft Windows 11 Home Single Language 10.0.26100.3915 Build 26100.3915
* Using the virtualbox driver based on user configuration
* Downloading VM boot image ...
E0515 22:20:40.874004 23116 iso.go:90] Unable to download https://storage.googleapis.com/minikube/iso/minikube
/iso/minikube-v1.35.0-amd64.iso?checksum=file:https://storage.googleapis.com/minikube/iso/minikube-v1.35.0-amd64
: Mode:2 Umask:----- Detectors:[0x633f340 0x633f340 0x633f340 0x633f340 0x633f340 0x633f340] Deco
00615820 tar.xz:0xc000615830 tar.zst:0xc000615840 tbz2:0xc000615810 tgz:0xc000615820 txz:0xc000615830 tzst:0xc00
c0007866e0 https:0xc000786730] Dir:false ProgressListener:0x62e2040 Insecure:false DisableSymlinks:false Options
s.com/minikube/iso/minikube-v1.35.0-amd64.iso.sha256": dial tcp: lookup storage.googleapis.com: no such host
* Downloading VM boot image ...
```

```
C:\Users\priya>minikube start --no-vtx-check
* minikube v1.35.0 on Microsoft Windows 11 Home Single Language 10.0.26100.3915 Build 26100.3915
* Automatically selected the virtualbox driver
* Downloading VM boot image ...
50515-22-20-21-144460-6160-iso-gz-00] Unable to download https://storage.googleapis.com/minikube/iso
```

Check in virtual box



```
C:\Users\DYPATU>minikube config set driver virtualbox
```

```
C:\Users\DYPATU>minikube status
```

```
C:\Users\DYPATU>kubectl get nodes
```

```
C:\Users\DYPATU>kubectl get po -A
```

```
C:\Windows\System32>minikube config set driver virtualbox
! These changes will take effect upon a minikube delete and then a minikube start
```

```
C:\Windows\System32>minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

```
C:\Windows\System32>kubectl get nodes
NAME        STATUS    ROLES    AGE      VERSION
minikube    Ready     control-plane  6m14s    v1.32.0
```

```
C:\Windows\System32>kubectl get po -A
NAMESPACE   NAME                                     READY   STATUS    RESTARTS   AGE
kube-system  coredns-668d6bf9bc-x9nhm              1/1     Running   0           14m
kube-system  etcd-minikube                          1/1     Running   0           14m
kube-system  kube-apiserver-minikube                1/1     Running   0           14m
kube-system  kube-controller-manager-minikube        1/1     Running   0           14m
kube-system  kube-proxy-k4279                       1/1     Running   0           14m
kube-system  kube-scheduler-minikube                 1/1     Running   0           14m
kube-system  storage-provisioner                    1/1     Running   1 (13m ago) 14m
```

```
C:\Users\DYPATU>kubectl create deployment hello-minikube --image=kicbase/echo-server:1.0
deployment.apps/hello-minikube created
```

```
C:\Users\DYPATU>kubectl get pods
```

```
C:\Windows\System32>kubectl create deployment hello-minikube --image="kicbase/echo-server:1.0"
deployment.apps/hello-minikube created
```

```
C:\Windows\System32>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
hello-minikube-ffcbb5874-p7cr9     1/1     Running   0           15s
```

```
C:\Users\DYPATU>kubectl expose deployment hello-minikube --type=NodePort --port=8080
```

```
C:\Users\DYPATU>kubectl get pods
```

```
C:\Users\DYPATU>kubectl get services hello-minikube
```

```
C:\Users\DYPATU>minikube service hello-minikube --url
```

```
C:\Windows\System32>kubectl run hello-minikube --image=kicbase/echo-server:1.0 --port=8080
pod/hello-minikube created
```

```
C:\Windows\System32>kubectl expose pod hello-minikube --type=NodePort --port=8080
service/hello-minikube exposed
```

```
C:\Windows\System32>kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
hello-minikube	1/1	Running	0	87s
hello-minikube-ffcbb5874-p7cr9	1/1	Running	0	20m

```
C:\Windows\System32>kubectl get services hello-minikube
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
hello-minikube	NodePort	10.99.94.185	<none>	8080:30603/TCP	88s

```
C:\Windows\System32>minikube service hello-minikube -url
```

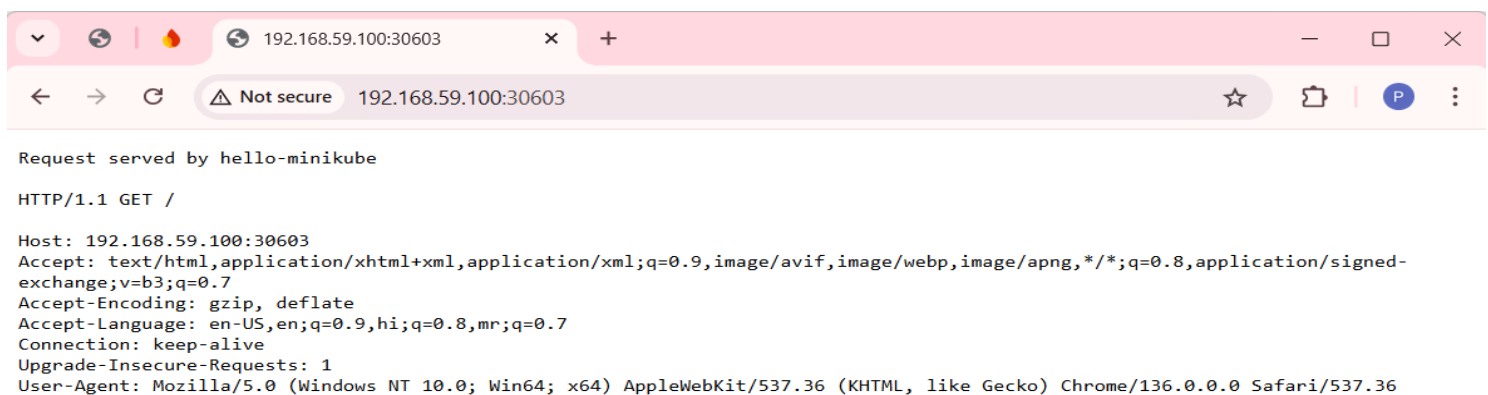
```
Error: unknown shorthand flag: 'u' in -url
```

```
See 'minikube service --help' for usage.
```

```
C:\Windows\System32>minikube service hello-minikube --url
```

```
http://192.168.59.100:30603
```

Check in browser using given local host address



C:\Users\DYPATU>kubectl run nginx --image=nginx

C:\Users\DYPATU>kubectl get pods

```
Command Prompt
Microsoft Windows [Version 10.0.26100.3915]
(c) Microsoft Corporation. All rights reserved.

C:\Users\priya>kubectl run nginx --image=nginx
pod/nginx created

C:\Users\priya>kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
hello-minikube	1/1	Running	0	8m34s
hello-minikube-ffcbb5874-p7cr9	1/1	Running	0	27m
nginx	0/1	ContainerCreating	0	13s

Open new command prompt (cmd)

C:\Users\DYPATU>kubectl get deployments

C:\Users\DYPATU>kubectl get service

```
Command Prompt
Microsoft Windows [Version 10.0.26100.3915]
(c) Microsoft Corporation. All rights reserved.

C:\Users\priya>kubectl get deployments
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
hello-minikube	1/1	1	1	28m

```
C:\Users\priya>kubectl get service
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
hello-minikube	NodePort	10.99.94.185	<none>	8080:30603/TCP	9m3s
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	55m

C:\Users\DYPATU>kubectl delete service hello-minikube

C:\Users\DYPATU>kubectl get service

C:\Users\DYPATU>kubectl delete deployments hello-minikube

C:\Users\DYPATU>kubectl get deployments

```
C:\Users\priya>kubectl delete service hello-minikube
service "hello-minikube" deleted
```

```
C:\Users\priya>kubectl get service
NAME          TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
kubernetes    ClusterIP     10.96.0.1     <none>         443/TCP    56m
```

```
C:\Users\priya>kubectl delete deployments hello-minikube
deployment.apps "hello-minikube" deleted
```

```
C:\Users\priya>kubectl get deployments
No resources found in default namespace.
```

C:\Users\DYPATU>kubectl get pods

C:\Users\DYPATU>minikube stop

```
C:\Users\priya>kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
hello-minikube 1/1     Running   0           14m
nginx          1/1     Running   0           6m23s
```

```
C:\Users\priya>minikube stop
E0516 04:56:15.366496 15148 daemonize_windows.go:38] error terminating scheduled
schedule-stop service for profile minikube: NewSession: new client: new client: Error
using: docker, &{[] [C:\Users\priya\.minikube\machines\minikube\id_rsa]}: open C:\
be\id_rsa: Access is denied.
* Stopping node "minikube" ...
* 1 node stopped.
```

C:\Users\DYPATU>minikube start

```
C:\Users\priya>minikube start
* minikube v1.35.0 on Microsoft Windows 11 Home Single Language 10.0.26100.3915
* Using the virtualbox driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Restarting existing virtualbox VM for "minikube" ...
```

